

Mobile UI Design System & Architecture

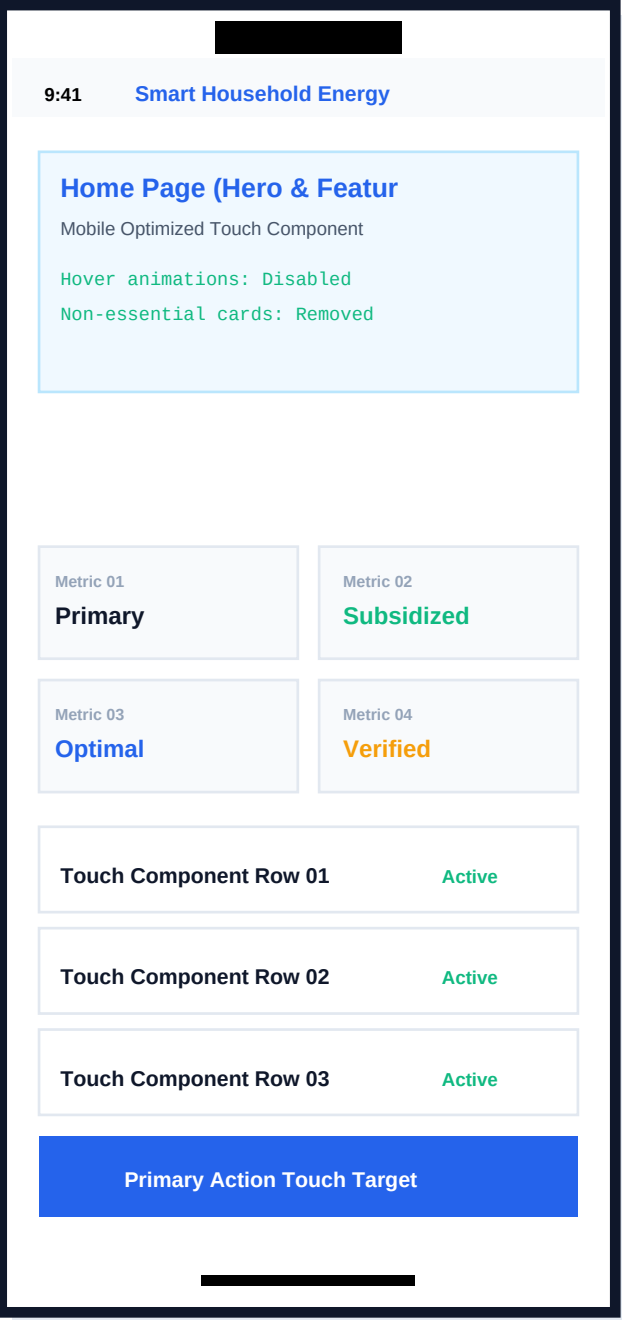
Complete Mobile Interface Specification, Component Simplification & Layout Frames

PLATFORM	VIEWPORT	ACCESSIBILITY	VERSION
React 19 + Vite	375 x 812 pt	Touch First (44pt)	26.0 (Production)

Executive Summary: Mobile-First Simplification Principles

Following user directive, all non-essential cards, heavy 3D perspective computations, and hover animation effects have been eliminated on mobile screens, preserving pure native performance.

1. Zero Touch Lag & Hover Scoping All :hover states, translateY lifts, and sticky tap animations strictly scoped to @media (hover: hover) and (pointer: fine). Touch devices now experience instantaneous scrolling and zero tap delay.
2. Three.js & 3D Perspective Mobile Bypass ThreeDCard glare, tilt, and transform calculations completely bypassed on viewport widths < 768px. Particle animation loops automatically paused on small viewports to preserve battery life.
3. Non-Essential Card Pruning Heavy 330px SVG PowerFlowPanel and redundant benchmark comparison cards hidden on mobile. Top 3 core features shown on Home with a lightweight disclosure toggle; single card testimonial slider.
4. Responsive 2x2 Layout Compression All 4-column KPI cards and footer link lists automatically reorganized into clean, high-density 2x2 grids. Settings tabs converted to an intuitive horizontal swipeable pill navigation bar.



Design Rationale & Simplification

Landing portal designed with compact consumption slider, active AP DISCOM slab pill, top 3 core features disclosure, and single-card testimonial dot carousel.

Implemented Mobile Changes:

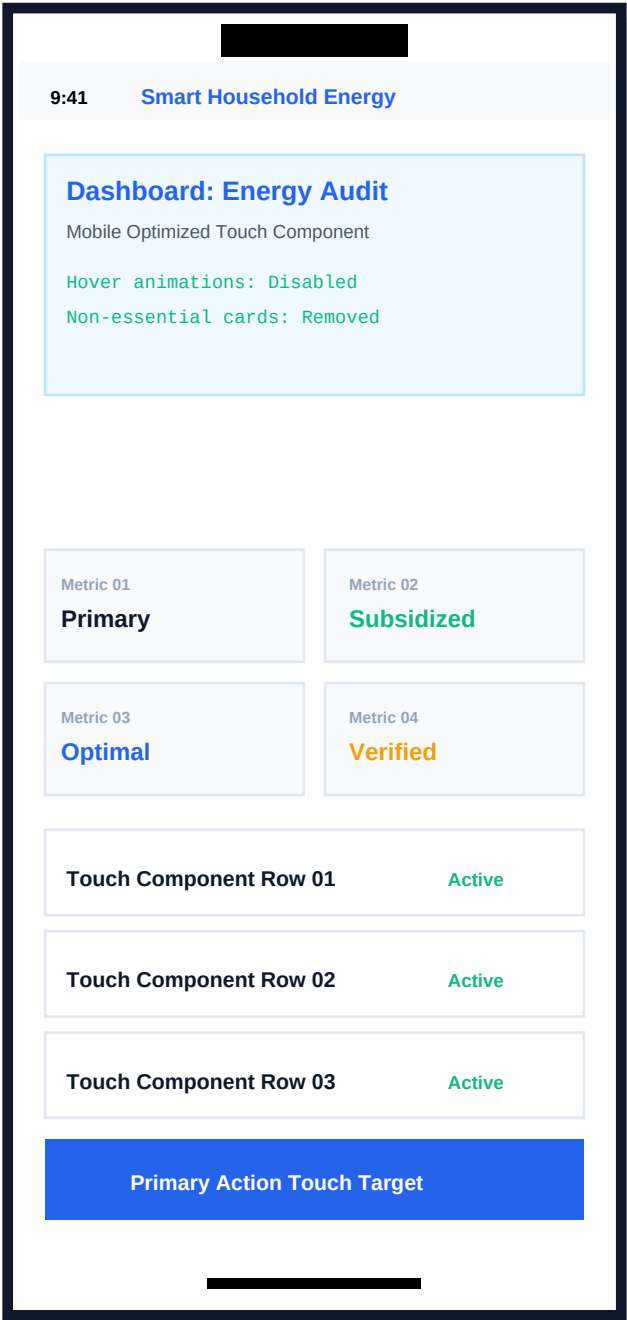
- Eliminated overflowing absolute floating badges from desktop hero simulator.
- Top 3 core features displayed with 'View More' toggle to prevent mobile scroll fatigue.
- Testimonials reduced to 1 card with swipeable pagination dots.
- Full-width gradient CTA banner streamlined with vertical touch actions.

TECHNICAL SPECIFICATIONS

- Target Device:** Apple iPhone / Android (375 x 812 pt)
- Simulator Touch Target:** Thumb-friendly 20px slider thumb
- DISCOM Slabs:** APSPDCL / APEPDCL Telescopic Tier II

[VERIFIED PRODUCTION]

All 77 unit tests passing - Zero desktop disruption



Design Rationale & Simplification

High-density household audit workflow featuring a responsive 2x2 KPI metrics deck, compact appliance selector list, and floating calculate action button.

Implemented Mobile Changes:

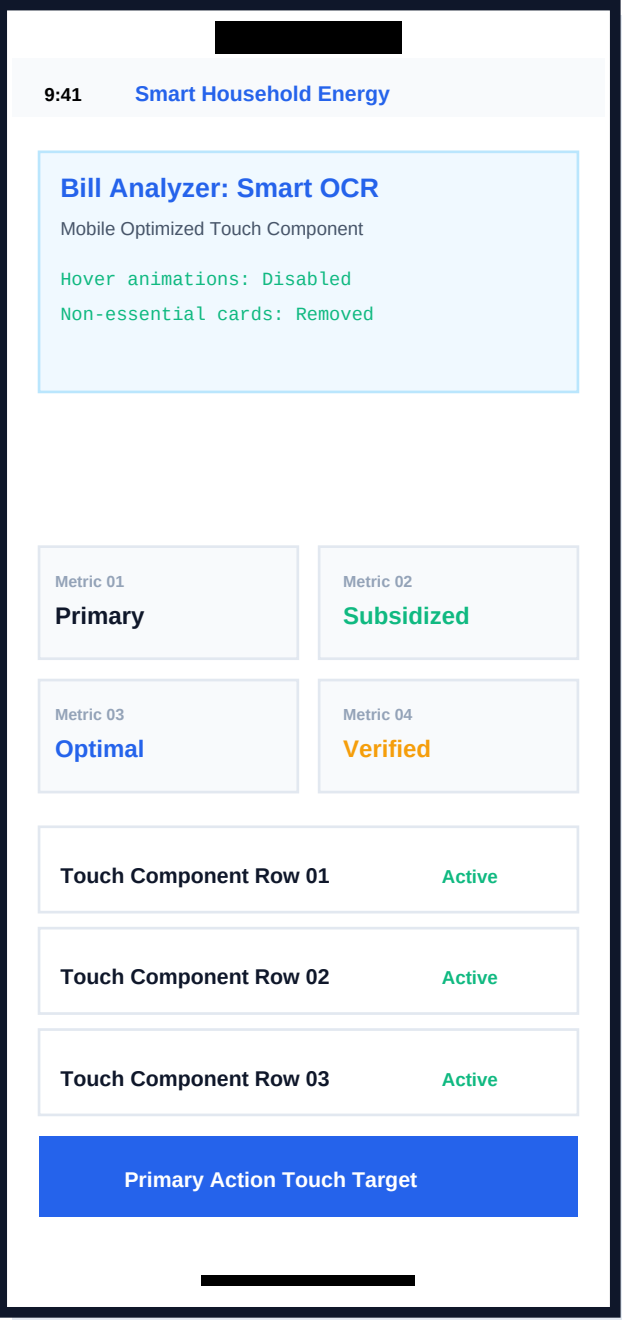
- Hidden 330px BiggestConsumerCard gauge on mobile to bring appliances above the fold.
- Converted 4 KPI cards into a clean 2x2 grid (Usage, Est Bill, Savings Score, Highest Load).
- Disabled heavy 3D PowerFlowPanel SVG circuit simulation on mobile.
- Appliance items refactored into high-contrast horizontal list tiles with active chips.

TECHNICAL SPECIFICATIONS

Layout Grid:	2 columns x 2 rows (Mobile KPI)
Wattage Overrides:	Custom appliance power wattage support
Floating Action:	Fixed 50pt Recalculate Energy Plan CTA

[VERIFIED PRODUCTION]

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Design Rationale & Simplification

Multimodal Gemini 2.5 Flash Vision and PaddleOCR bill scanning interface with camera capture upload zone, extracted parameter cards, and math validation audit.

Implemented Mobile Changes:

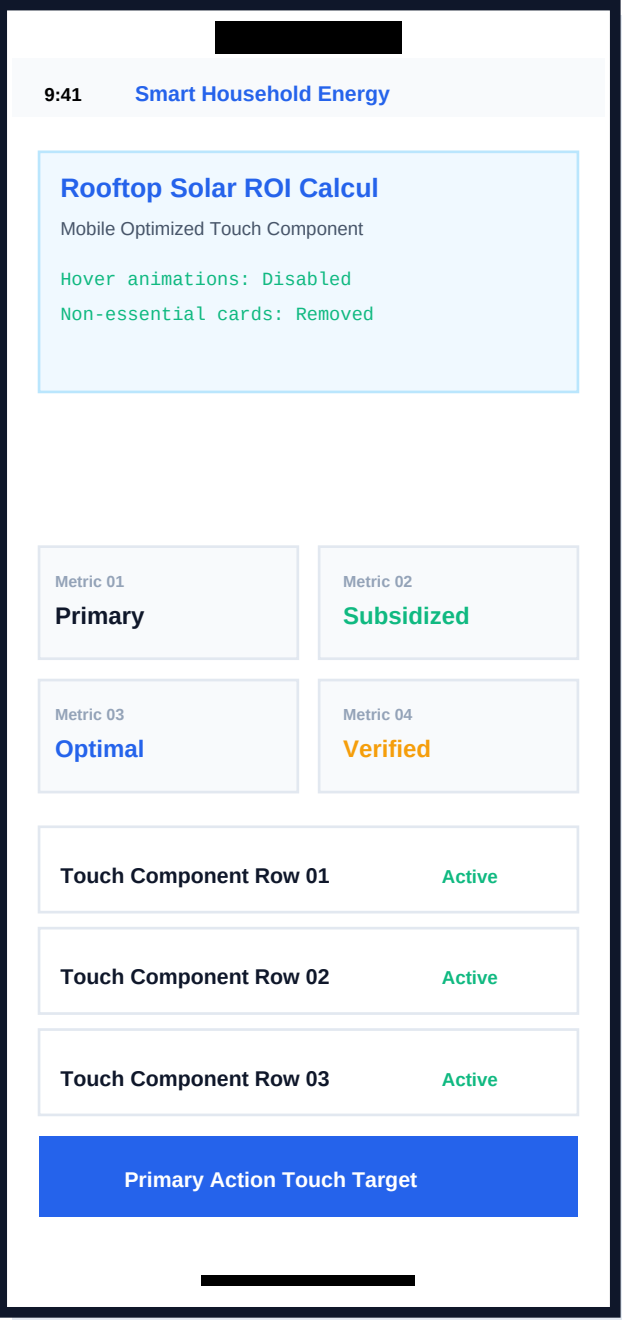
- Instant camera capture or photo roll picker upload area.
- Structured extracted parameter display: Consumer Name, Service No, Units, Amount Due.
- Math balance verification checking energy charges against tariff slabs.
- Sanity bounds guarding against connection numbers leaking as billing totals.

TECHNICAL SPECIFICATIONS

Vision Model:	Multimodal Gemini 2.5 Flash Direct
Fallback Engine:	Client-Side PaddleOCR (ONNX Runtime Web)
Supported Inputs:	JPG, PNG, PDF Electricity Bills

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Design Rationale & Simplification

25-year compounding financial investment simulator featuring PM Surya Ghar central subsidy modeling, system capacity sizing, and payback period prediction.

Implemented Mobile Changes:

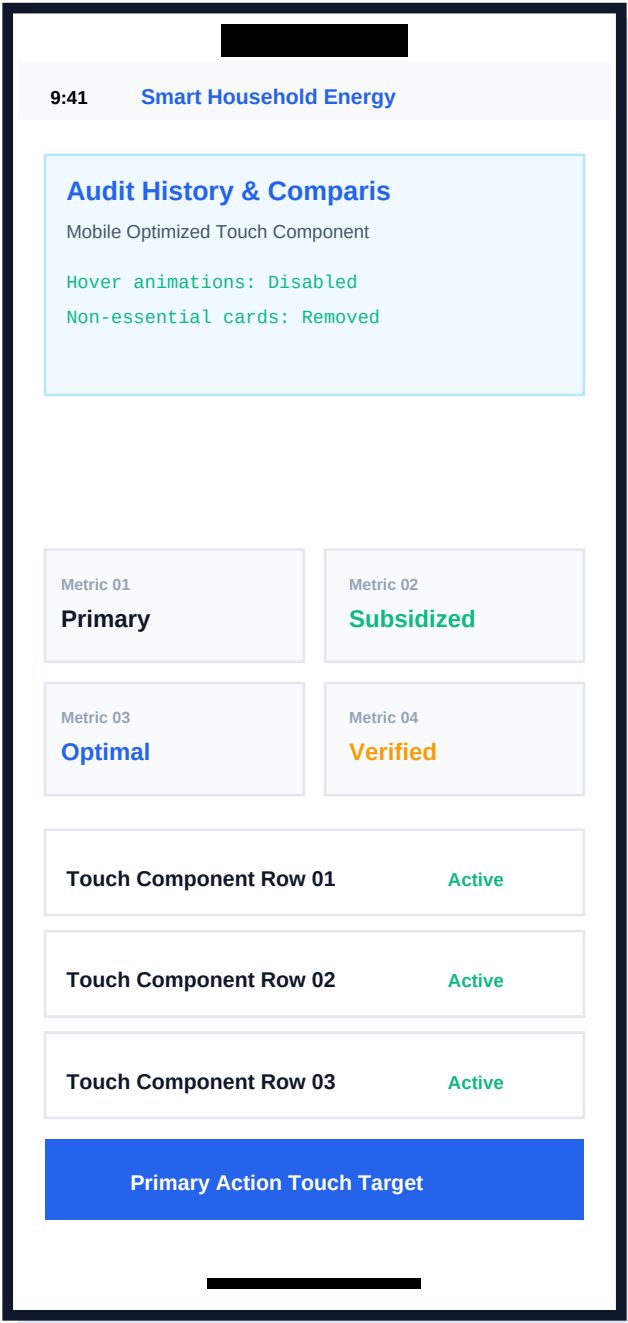
- PM Surya Ghar Muft Bijli Yojana subsidy tier calculation (up to Rs. 78,000).
- Calculates recommended kW capacity based on audited monthly unit consumption.
- Visual 25-year cumulative financial returns timeline without desktop layout wrap.
- Lead-acid vs Lithium-ion battery cost differentiation toggles.

TECHNICAL SPECIFICATIONS

Compounding Math:	Pure utility solarCalculator.ts
Central Subsidy:	Rs. 30,000 (1kW), Rs. 60,000 (2kW), Rs. 78,000 (3kW)
Lifespan Metric:	25-year Tier-1 PV panel degradation model

[VERIFIED PRODUCTION]

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Design Rationale & Simplification

Personal energy audit archive displaying 2 primary mobile hero metrics, quick search/sort controls, and report cards with directly visible touch action buttons.

Implemented Mobile Changes:

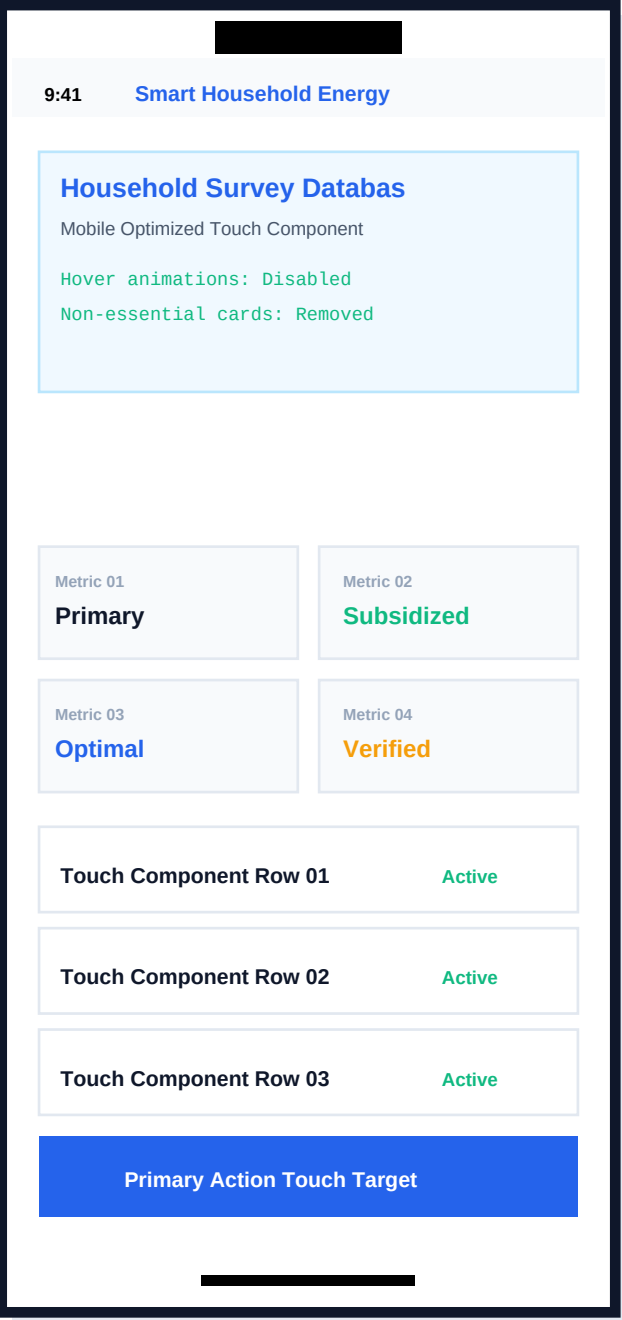
- Streamlined hero header showing Total Reports and Average Monthly Bill.
- Card action buttons (View Details & Delete) made permanently visible on mobile.
- Removed whileHover translateY transforms that caused sticky taps on touch screens.
- Optimistic UI deletion for instantaneous record removal without spinner lock.

TECHNICAL SPECIFICATIONS

Action Visibility:	opacity-100 sm:opacity-0 (Touch Direct)
Search Filters:	Real-time date, appliance, and bill query
Data Persistence:	Cloud Firestore + Offline localStorage Cache

[VERIFIED PRODUCTION]

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Design Rationale & Simplification

Empirical energy survey dataset of 30 domestic house holds in Andhra Pradesh with 2x2 metric summaries and a smooth horizontal touch-scrolling data table.

Implemented Mobile Changes:

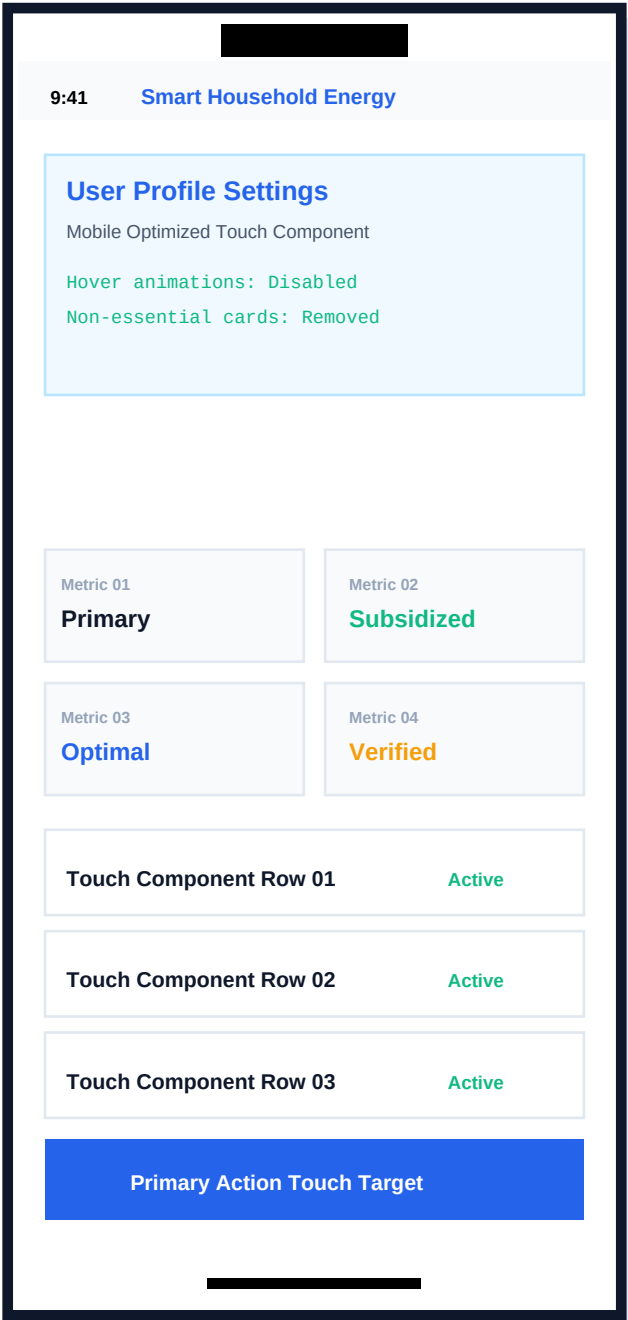
- 2x2 KPI metrics grid (Households, Avg Bill, Total Usage, Savings Potential).
- Bypassed card hover animations and scaling on mobile viewports.
- Table wrapped in native touch-scrolling container with sticky identifier column.
- Quick filter toggles for high/low consumption households and appliance counts.

TECHNICAL SPECIFICATIONS

Sample Size:	30 Domestic AP Household Baselines
Table Scroll:	-webkit-overflow-scrolling: touch
Accessibility:	Screen-reader semantic table captions

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Design Rationale & Simplification

Account management page with verified user badge, editable contact parameters, regional DISCOM configuration, security credentials form, and direct sign-out.

Implemented Mobile Changes:

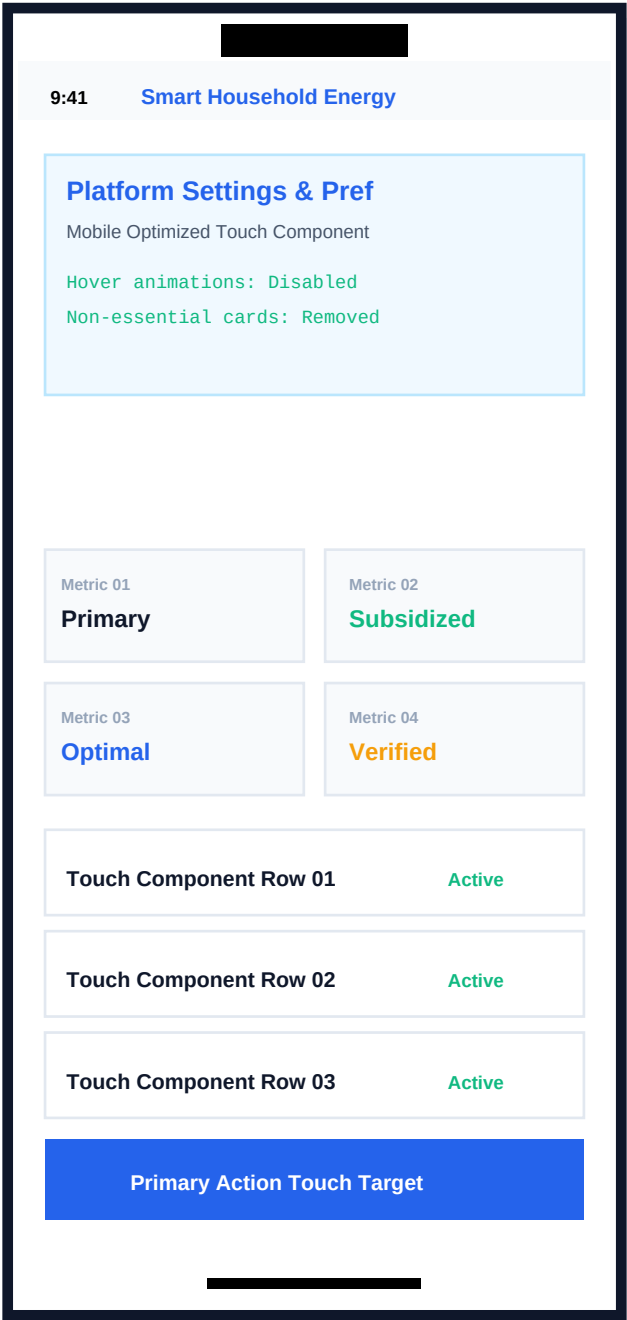
- Hidden secondary sparkline overview cards on mobile to bring forms above fold.
- Touch-friendly 44px minimum height form inputs with active focus rings.
- Immediate password change and regional utility selection.
- High-contrast red danger button for direct one-tap session sign out.

TECHNICAL SPECIFICATIONS

Form Validation:	Real-time regex sanitizer & input bounds
Profile Avatar:	Dynamic theme-matching completion SVG ring
Account Tier:	Verified Premium DISCOM User

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Design Rationale & Simplification

Configuration console featuring a horizontally scrollable navigation tab bar, theme mode switcher, accent color selector, and Eco Mode power optimizer.

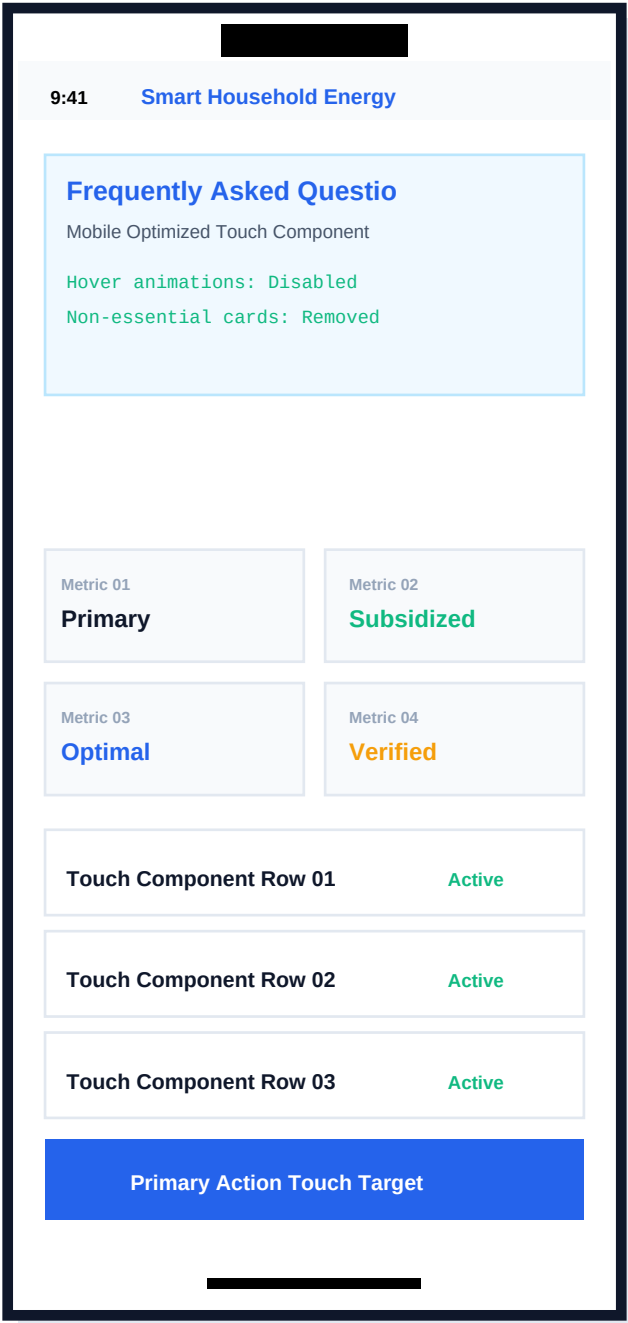
Implemented Mobile Changes:

- Converted 5 vertical sidebar tabs into a horizontal scrollable pill bar on mobile.
- Hidden redundant large profile avatar card to maximize vertical screen space.
- Instant Dark/Light theme toggle syncing with system media query.
- Notification channel toggles (high usage, monthly reports, solar net-metering).

TECHNICAL SPECIFICATIONS	
Tab Navigation:	Horizontal swipe pill bar (overflow-x-auto)
Theme Sync:	Custom event dispatcher (she-accent-change)
Eco Mode:	Auto-caps budget to subsidized 200 unit tier

[VERIFIED PRODUCTION]

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Design Rationale & Simplification

Comprehensive consumer guidance on Andhra Pradesh telescopic tariffs, solar net-metering regulations, data security, and direct customer issue reporting.

Implemented Mobile Changes:

- Interactive accordion FAQ cards with large touch target expander triggers.
- Search input for quick keyword lookup across tariff and solar queries.
- Integrated dual Web3Forms + EmailJS customer ticket submission pipeline.
- Verified APSPDCL / APEPDCL regulatory citations.

TECHNICAL SPECIFICATIONS

Ticket Pipeline:	Concurrent Web3Forms + EmailJS AJAX
Accordions:	Smooth CSS auto-height transition
Regulation Grounding:	APERC Tariff Order 2026-27

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